IDEATHON

MediScribe Ai

- **Problem Statement Title - Ineffieincy of medical data for doctors**
- **Theme – Health and BioTechnology**
- **Team Name – MediScribe Ai**

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PROBLEM IDENTIFIED

❖Selected Theme /Problem (Describe your Problem)

- Detailed explanation of the proposed problem
- Medical data remains fragmented across disparate formats like handwritten notes and PDFs, causing critical information loss and cognitive overload for doctors. Our RAG-powered journal centralizes these multimodal inputs into an AI-driven management system to enable instant, grounded clinical retrieval.
- Literature Gap
- It is based on an idea that doctors have no platform that is digitalized and dedicated to improving all requirements and needs of a doctor from data analysis to scheduling appointments
- Need of Addressing the Problem (Desirability)
- Every second counts for a medical professional as well as for patient hence efficiency in research and basic tasks that they need to perform will boost the performance that might save time and lives

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IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- It is a platform that allows medical professionals to re-engineer management of all and any type of data completely by moving from manually maintained data to a digitalised platform and we have provided all the neccessary tools that a medical professional must require based on research.
- How it addresses the problem
- We have created a RAG integrated application which will have all the neccessary tools to improve productivity for all medical professionals.
- Innovation and uniqueness of the solution
- We provide a platform that does not use Ai or LLM for our product. It is essentially only takes the data that the medical professionals have uploaded and provides answers based on that content without generating or hallucinating answers based on a LLM.
- We also integrate all possilble features that might be needed for a medical professional rather than only providing some of them.
- Doctors can access data without the requirement of an internet connection

TECHNICAL APPROACH

Technologies to be used

- **Frontend:** React 18 (Vite), TypeScript, Tailwind CSS, Framer Motion.
- **Backend:** Node.js, Express.js, SQLite (MVP) / PostgreSQL (Production).
- **AI Core:** Google Gemini Pro (Reasoning), Gemini 1.5 Flash (Vision & Audio).
- **Security:** AES-256 Encryption for Patient Health Information (PHI).

Methodology and process for implementation

Input Layer

Clinician uploads PDF, speaks (Voice), or draws (Canvas).

↓

Processing Layer (Gemini AI + RAG)

Voice → Transcribed to Text.
PDF → Vectorized for Contextual Search.
Sketch → Analyzed for Medical Context.

↓

Output Layer

Structured Patient Record saved to Secure Database.

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FEASIBILITY AND VIABILITY

Analysis of feasibility

- **High Feasibility:** MVP is fully functional with real-time data persistence and AI integration.
- **Scalable Architecture:** Decoupled Frontend/Backend allows easy migration to Cloud (AWS/GCP).

Potential challenges and risks

- **Data Privacy:** Strict handling of sensitive Patient Health Information (PHI).
- **AI Latency:** Processing large medical PDFs in real-time.
- **Adoption:** Steep learning curve for non-tech-savvy doctors.

Strategies for overcoming challenges

- **Privacy:** HIPAA-eligible cloud providers (GCP) + Role-Based Access Control (RBAC).
- **Performance:** Intelligent caching and 'Gemini Flash' models for sub-second responses.
- **Adoption:** Apple-inspired intuitive UI to lower entry barriers.

THE LINK FOR MVP: https://ai.studio/apps/dbc6010e-037e-4e98-ac95-86ce86958116?fullscreenApplet=true

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IMPACT AND BENEFITS

Potential impact on target audience

- **Better Decisions:** AI-assisted retrieval ensures no critical data is missed.
- **Work-Life Balance**: Automating notes reduces burnout.
- **Professional Growth:** Integrated research tools help doctors stay updated.

Benefits of the solution

Social: Restores the doctor-patient relationship by removing screen distractions.

Economic: Optimizes clinic workflow, allowing for higher patient throughput.

Operational: Cuts data retrieval time by half; ensures secure, organized storage.

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MONETIZATION STRATEGY

Revenue sources

- **Freemium (Community Building):** Free basic access for medical students to build long-term loyalty.
- **SaaS Subscription (B2C):** 'Professional Tier' for individual doctors ($/month) offering unlimited AI queries.
- **Enterprise Licensing (B2B):** Custom contracts for Hospitals/Clinics including EHR integration.

Cost structure

- **API Usage:** Google Gemini Token consumption (Variable).
- **Infrastructure:** Cloud hosting (AWS/GCP) and Database storage.
- **Development:** Continuous security updates and AI model fine-tuning.

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COST ESTIMATION

Resources Required

- Manpower: 2 Full Stack Developers, 1 UI/UX Designer, 1 AI Engineer.
- Equipment: High-performance workstations, Tablets/iPads for testing 'Dr. Doodle'.

Tech Stack Budget (Initial Phase)

- Google Gemini API Credits: ~$500/month (Pilot Phase).
- Cloud Hosting (Vercel/Render): ~$50/month.
- Domain & SSL Security: ~$20/year.

Total MVP Budget: Low capital requirement (Software-focused development).

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RESEARCH AND REFERENCES

Details / Links of the reference

- Burnout Statistics: “Physician Burnout: The Root of the Problem” (National Academy of Medicine).
- RAG Architecture: “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (Lewis et al., 2020).
- Technical Docs: Google Gemini API Documentation & React.js Documentation.
- Design Philosophy: Apple Human Interface Guidelines (Design for Clarity & Deference).

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FRONTEND ARCHITECTURE: A CRAFTED EXPERIENCE

Core Framework: React 18 + Vite

- Selected for component-based reusability and optimized production builds.

Design System

- **Tailwind CSS**: Utility-first framework for rapid, consistent medical-grade styling.
- **Framer Motion**: Handles fluid transitions (modal scaling, slide-overs) to reduce cognitive load.
- **Lucide React**: Provides consistent, clean iconography.

Architecture

- **Single Page Application (SPA)**: Ensures seamless user interaction without page reloads.
- **Direct AI Integration**: Client-side handling of non-sensitive AI interactions for responsiveness.

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BACKEND & AI CORE: SECURE & SCALABLE

Backend API (Express.js)

- **Lightweight**: Handles CRUD operations for journals and patient history.
- **Modular**: Designed to swap data layers (Local -> Cloud) without rewriting logic.

Data Persistence

- **Current**: SQLite (Local file-based) for rapid MVP prototyping.
- **Roadmap**: Migration to PostgreSQL / Google Cloud Firestore for concurrent user scaling.

AI Integration Strategy

- **Decoupled**: Independent scaling of AI services from the core application.
- **Security**: Server-side validation of all API requests to protect tokens.

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TECHNICAL Q&A & FUTURE FUTURE ROADMAP

Q: How does the system scale with a file-based database?

A: The backend features a modular data access layer. Migration to PostgreSQL is a planned configuration change, not a rewrite.

Q: How do you prevent medical AI medical AI hallucinations?

A: Strict RAG Enforcement. The system prompt explicitly forbids external knowledge, forcing the AI to answer only from the uploaded PDF context.

Q: What is the strategy for HIPAA compliance? for HIPAA compliance?

A: Production will utilize HIPAA-eligible Google Cloud services, BAA signing, encryption at rest/transit, and strict Role-Based Access Control (RBAC).

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